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Time Crystals: From a No-Go Theorem to Driven Phases of Matter Soumyabroto Majumder,1 Rahul Chandra,2 Analabha Roy,1 and Mahbub Rahaman3, ∗ 1Department of Physics, The University of Burdwan, Burdwan 713104, India 2Sankara Nethralaya Elite School of Optometry, DJ/16, Action Area 1D, New Town, Rajarhat, Kolkata 700156, India 3Harish Chandra Research Institute, A CI of Homi Bhabha National Institute, Chhatnag Road, Jhunsi, Prayagraj, Uttar Pradesh 211019, India Spontaneous symmetry breaking is fundamental to the emergence of spatial order in condensed matter and occurs when a ground or thermal state possesses less symmetry than its governing Hamiltonian.

Wilczek proposed an analogous breaking of time translation symmetry in which a quantum system in its lowest energy state would exhibit persistent periodic motion.

Although this equilibrium proposal was subsequently challenged, the concept of time crystalline order was reformulated in nonequilibrium systems.

In periodically driven systems, continuous time translation symmetry is reduced to a discrete symmetry, whose spontaneous breaking can give rise to a discrete time crystal (DTC) characterized by robust oscillations with a period equal to an integer multiple of the driving period.

Because generic driven systems absorb energy and eventually heat toward infinite temperature, the emergence of such order requires mechanisms, including many-body localization, prethermalization, or integrability that suppress heating.

We review the Floquet algebraic origin of long lived oscillations and examine their dependence on driving protocols, initial states, and interactions, while highlighting theoretical and experimental advances and the roles of symmetry, dimensionality, and dissipation in stabilizing DTCs.

Keywords: time crystals; Floquet systems; integrability; dynamical symmetries I.

INTRODUCTION level of one-point observables, time-translation symmetry breaking (TTSB), like conventional spontaneous sym-Symmetry is a central organizing principle of nature.

metry breaking, must be identified through correlation In condensed matter, the spontaneous breaking of sym-functions.

Watanabe and Oshikawa therefore character-metry underlies the emergence of order.

A ferromagnet ized TTSB using the correlator of an extensive observable ˆ R d ˆ breaks spin rotation symmetry, while a crystal breaks O = d x O(x), continuous spatial translations down to a discrete lattice 1 iHtˆ ˆ −iHtˆ ˆ subgroup.

In both cases, the ground state or thermal state f(t) = lim 2 e O e O , (1) exhibits less symmetry than the Hamiltonian from which V →∞ V it arises.

The search for new phases of matter, and for the and showed that f(t) is time independent in the ground symmetries they break, is central to the development of state and in Gibbs states of local Hamiltonians.

The Gibbs condensed matter physics and to technological progress.

state result was later established rigorously (Watanabe Since relativity places space and time on an equal footing, et al., 2020).

Time crystals are therefore forbidden in it is natural to ask whether time-translation symmetry equilibrium.

can be broken in the same way.

The idea of TTSB was revived by considering quan-In 2012, Wilczek and Shapere posed this question tum systems driven far from equilibrium.

In a periodi-for quantum (Wilczek, 2012) and classical (Shapere & cally driven many-body quantum system, the Hamiltonian Wilczek, 2012) systems, respectively, and introduced the satisfies Hˆ (t + T) = Hˆ (t), and only the discrete time-term time crystal.

Their proposal envisioned a quantum translation symmetry t → t + T remains.

If the state system in its lowest energy state undergoing perpetual and the corresponding observables respond with a period periodic motion, such as a rotating density modulation mT , where m ≥ 1 is an integer, this discrete symme-of interacting charged bosons on a ring threaded by mag-try is spontaneously broken.

This is the essence of a netic flux.

Although this motion would not permit energy discrete time crystal (DTC) (Else et al., 2016; Khemani extraction, it appeared inconsistent with the conventional et al., 2016; Sacha, 2015; von Keyserlingk et al., 2016; Yao understanding that equilibrium states are stationary.

The et al., 2017).

The response must also be robust against proposal was soon challenged: Bruno(2013) showed that perturbations, accompanied

by spatial long-range order, the persistent motion envisaged in Wilczek's ring model and persist in the thermodynamic limit.
The main ob-cannot occur in a ground state, while Watanabe and Os-stacle is that generic interacting driven systems absorb hikawa(2015) established a general no-go theorem.
Since energy and heat to a featureless infinite-temperature state.
every eigenstate and Gibbs state is stationary at the The first proposals avoided this through disorder-induced many-body localization, while prethermalization at high driving frequency and long-range interactions were later identified as disorder-free routes (Else et al., 2017; Pizzi ∗ Contact author: mahbub.phys@gmail.com et al., 2021; Russomanno et al., 2017).
Soon after these 2 proposals, signatures of period doubling were reported tions of the Schr¨odingerequation can be written as $-i\mu\alpha t$ in trapped ions (Zhang et al., 2017) and in dense en- $|\psi\alpha(t)\rangle = e\,|\phi\alpha(t)\rangle$, where $|\phi\alpha(t + T)\rangle = |\phi\alpha(t)\rangle$ and sembles of nitrogen-vacancy (NV) centers (Choi et al., the quasienergies $\mu\alpha$ are defined modulo ωD.
The Floquet 2017).
These were followed by observations in NMR sys-eigenstates $|\phi\alpha(0)\rangle$ are invariant under $U\hat{}F$ up to a phase.
tems (Pal et al., 2018; Rovny et al., 2018), nuclear-spin They therefore maintain the discrete time translation sym-registers (Randall et al., 2021), prethermal trapped-ion metry $t \rightarrow t+T$, and play the role that energy eigenstates chains (Kyprianidis et al., 2021), and superconducting and Gibbs states play in equilibrium.
Any breaking of quantum processors (Mi et al., 2022).
Comprehensive this symmetry must therefore be spontaneous.
reviews of the field are available (Else et al., 2020; Khe-Discrete time translation symmetry is spontaneously mani et al., 2019; Sacha & Zakrzewski, 2018; Zaletel et al., broken if, for generic short range correlated initial states 2023), together with a monograph (Sacha, 2020).
$|\psi(0)\rangle$ (for instance product states), the stroboscopic ex-In this short review, we focus on the Floquet algebraic pectation value $f(nT) = \langle\psi(nT)|O\hat{}\,\psi(nT)\rangle$ of some local origin of long-lived oscillations and their dependence on observable $O\hat{}$ is not T -periodic but repeats only after driving protocols, initial states, and interactions.
We $m \geq 1$ periods, highlight theoretical and experimental advances and the roles of symmetry, dimensionality, and dissipation in sta-$f(n + 1)T \neq f(nT)$, $f(n + m)T = f(nT)$.
(3) bilizing DTCs.
We restrict our attention to disorder-free spin-1/2 systems, both interacting and integrable, where This manifests as a subharmonic response, which can be many body localization is unavailable and robustness readily observed as a peak at frequency $\omega D/m$ in the must come from other mechanisms.
We also discuss the Fourier spectrum of $f(nT)$.
Equivalently, the Floquet role of integrability, which suppresses heating and can eigenstates are long range correlated cat states.
Their sustain persistent oscillations, and ask when these oscil-quasienergies are organized in multiplets split by $\omega D/m$, lations survive integrability breaking perturbations and which is the Floquet analogue of the correlator criterion so constitute genuine time crystalline order.
Section II in Eq.
(1) (Else et al., 2016; Khemani et al., 2017; von introduces the Floquet formalism, the criteria for time Keyserlingk et al., 2016).
crystalline order, and the main stabilization mechanisms.
A subharmonic response alone, however, does not guar-Section III examines integrability and the dynamical sym-antee a discrete time crystal.
The response must emerge metries that underlie persistent oscillations.
SectionIV from the collective dynamics of the interacting driven sys-reviews experimental realizations, and Sec.V discusses tem.
Non-interacting spins subjected to perfect π pulses open problems.
flip every period and trivially show period doubling.
A spin rotation error εr, however, shifts the response fre-quency from π/T to $\pi(1 - \varepsilon r)/T$, so the period is not II.
DISCRETE TIME CRYSTALS IN FLOQUET locked.
To label a system a discrete time crystal, it must SYSTEMS therefore satisfy the following criteria (Else et al., 2016; Khemani et al., 2016; Yao et al., 2017): A.
Floquet formalism and criteria for time-crystalline order a) The system should exhibit spontaneous breaking of discrete time translation symmetry: for generic short-range correlated initial states, the strobo-With the no-go theorem, the search for time

crystals scopic expectation value of a local observable should shifted to nonequilibrium systems.
The first proposals satisfy Eq.
(3), even though the Hamiltonian is pe-were for periodically driven (Floquet) systems, which riodic, Hˆ (t) = Hˆ (t + T).
retain only a discrete time translation symmetry.
To break this symmetry, the system must respond with a b) The response should be rigid: the subharmonic period mT that is an integer multiple m ≥ 1 of the drive oscillations should remain locked at period mT over period T .
To understand how such a response can arise, a finite range of parameters, including imperfect we first review the Floquet formalism.
pulses, without fine tuning the system parameters.
For a Hamiltonian with period T = 2π/ωD, Hˆ (t + T) = Hˆ (t), the wavefunction |ψ(t)⟩ obeys the Schr¨odinger c) The oscillating order parameter should be long- ˆ range correlated in space, and the oscillations should equation i∂t |ψ(t)⟩ = H(t) |ψ(t)⟩ (we set $\hbar$ = 1).
The stroboscopic evolution, which determines the observables persist indefinitely in the thermodynamic limit.
The at times t = nT , is generated by the Floquet operator limits must be taken in the order limnT →∞ limN→∞, since in any finite system tunnelling between the cat " # Z T state partners eventually destroys the oscillation.
−iHˆF T UˆF ≡ Uˆ(T, 0) = T exp −i Hˆ (t) dt = e , 0 (2) Therefore, the dynamics of a periodically driven inter- |ψ(nT)⟩ = Uˆ n |ψ(0)⟩ , acting system must support a robust subharmonic re-F sponse that persists in the thermodynamic limit.
In where T denotes time ordering and HˆF is the Flo-generic driven systems, however, energy absorption from quet Hamiltonian.
By Floquet's theorem, the solu-the drive leads to heating toward infinite temperature.
3 Because a periodic drive conserves no energy, the Flo-t quet eigenstates of a generic interacting system satisfy 2T ψ0 spin flip a Floquet version of the eigenstate thermalization hy- | localization ˆ ® pothesis (ETH) (D'Alessio et al., 2016; Deutsch, 1991; H2 ˆ Srednicki, 1994).
Local observables then relax to their UF π gT1 = 2 (1 ²) infinite-temperature values, which destroys the subhar-p − e ˆ r H1 i x monic response (D'Alessio & Rigol, 2014; Lazarides et al., o Xˆ = σˆ T Xˆ ψ d i 0 i 2014a).
A central challenge is therefore to identify mech- | 2 T Y anisms that suppress heating while stabilizing the sub- ® ˆ T Uˆ = harmonic dynamics.
One widely studied approach is to H2 2 F Uˆ F iHˆ T iHˆ T localize the many-body system throughout the drive cycle.
e− 2 2 e− 1 1 Other approaches slow down or prevent heating without ˆ H1 T1 localization.
ωD = 2π/T 0 ψ0 Several theoretically and experimentally established | ® mechanisms can provide such localization or otherwise suppress heating.
The most widely used approach is to FIG.
1. Schematic of the discrete time crystal drive protocol.
introduce strong disorder, which can produce many-body The system evolves under two Hamiltonians, Hˆ1 and Hˆ2, for localization (MBL) (Abanin et al., 2019; Basko et al., durations T1 and T2 = T − T1, respectively.
The first step, π P x Hˆ1 = (1 − εr) σˆ , implements an approximately global 2006; Pal & Huse, 2010).
An MBL system possesses an 2T1 j j extensive set of quasi-local integrals of motion (Huse et al., spin flip with spin-rotation error εr, while the second is gener-2014; Serbyn et al., 2013), and MBL has been proven in ated by an interacting local or quasi-local Hamiltonian.
Over one-dimensional spin chains under an assumption on level a finite range of εr, the resulting stroboscopic magnetization exhibits a subharmonic response locked at period 2T .
statistics (Imbrie, 2016).
MBL persists under periodic driving (Lazarides et al., 2015; Ponte et al., 2015).
Its quasi-local integrals of motion inhibit energy absorption special drive amplitudes and frequencies, which constrains from the drive and thus prevent heating to infinite temper-the dynamics and produces long-lived nonthermal behav-ature.
MBL is stable against weak perturbations within ior.
the isolated system.
It is, however, destroyed by coupling A further approach is to consider an integrable system to an environment, and its stability in higher dimensions, whose dynamics are constrained by an extensive set of con-with long-range interactions, and asymptotically even in served quantities.

Such a system relaxes to a generalized one dimension remains under active debate (De Roeck & Gibbs ensemble rather than to infinite temperature.
Un- ˇ Huveneers, 2017; Morningstar et al., 2022; Suntajs et al., der appropriate conditions, integrability can then support 2020). persistent oscillations.
These arise when the dynamics A second approach is to use a clean, disorder-free admits a dynamical symmetry, an operator Aˆ satisfying system driven at high frequency.
When the drive fre- ˆ † ˆ ˆ iθ ˆ UF A UF = e A, for which θ = π gives period doubling.
quency is large compared with the local energy scale Jloc, In isolated Hamiltonian systems, the analogous condition heating is exponentially slow, with a rate bounded by [H,ˆ Aˆ] = ωAˆ sustains undamped oscillations (Buˇcaet al., −c ω /J e D loc (Abanin et al., 2017; Kuwahara et al., 2016; 2019; Medenjak et al., 2020).
For the kicked Ising chain Mori et al., 2016).
The system then enters a prethermal ˆ z of Sec.
IIB at perfect π pulses, A = σˆj satisfies this con-regime in which its dynamics are governed, for long but dition with θ = π on every site.
Each of the mechanisms finite times, by an effective time-independent Hamilto-above can then be viewed as a way of ensuring that a nian before eventual heating occurs.
By the Watanabe-z dressed version of σˆj survives pulse errors.
We return to Oshikawa theorem, such a static effective Hamiltonian dynamical symmetries in Sec.
IIIA. alone cannot host a time crystal.
For a drive containing a ˆ Section IIB introduces the generic drive protocol for near-π global spin flip P , however, the Floquet operator a discrete time crystal, shown schematically in Fig.1, ˆ ˆ −iDTˆ takes the form UF ≈ P e , where the effective Hamil-and reviews established results on the mechanisms that ˆ ˆ ˆ tonian D has an emergent Z2 symmetry, [D, P] = 0 (Else produce robust discrete time crystalline order.
et al., 2017).
If the prethermal state of Dˆ spontaneously breaks this symmetry, the system shows period doubling.
This requires a finite-temperature ordered phase, and the B.
Drive protocols and stabilization mechanisms oscillation persists up to the exponentially long heating time rather than strictly forever.
The canonical protocol for realizing a discrete time Additionally, precise engineering of the drive protocol crystal is based on a binary Floquet drive, shown schemat-can strongly suppress thermalization.
At special drive ically in Fig.1.
During each period, the system evolves parameters, two-tone drive protocols manifest exactly flat under two Hamiltonians, Hˆ1 and Hˆ2, for durations T1 Floquet bands and thereby suppress heating (Banerjee and T2 = T − T1, respectively.
Defining et al., 2025).
More generally, the leading-order Floquet −iHˆ1T1 −iHˆ2T2 Hamiltonian can acquire emergent conservation laws at Uˆ1 = e , Uˆ2 = e , (4) 4 the one-period Floquet operator is ² = 0.1 (a) DTC: Ji 0 free: Ji = 0 ˆ ˆ −iH2T2 −iH1T1 1 UˆF = Uˆ2Uˆ1 = e e .
(5) ®) T The first step is chosen to implement an approximately n (0 z i ˆ σ global spin flip, while the second is generated by an in- teracting local or quasi-local Hamiltonian.
For a spin-1/2 1 − chain, a representative choice is (Yao et al., 2017) 0 4 8 12 16 20 24 n = t/T ˆ X x (b) H1 = g σˆi , drive ω i DTC D] (6)) X T free peak at ωD/2 ˆ z z z ┌ n H = J σˆ σˆ + h σˆ , (2 i i i+1 i i z i ˆ σ ² ω i [D T F F where Ji and hi denote random couplings and longitudinal fields.
In the limit T → 0 with gT fixed, the first 0 1 1 3 1 1 1 4 2 4 step becomes an instantaneous kick, and the protocol is ω/ωD known as the kicked Ising chain.
Experimentally, the global spin flip is implemented by a nominal π pulse, FIG.
2. (a) Stroboscopic magnetization M(nT), Eq.
(9), at which is generally imperfect.
We parametrize the pulse εr = 0.1 for interacting (Ji ≥ 0) and non-interacting (Ji = 0) imperfection by chains.
The interacting chain shows period doubling locked at 2T , while the non-interacting chain shows beating.
(b) Fourier π spectrum of M(nT).

With interactions the peak is pinned at g = (1 − εr) , (7) 2T1 ω = π/T ; without interactions it shifts to ω = π(1 − εr)/T .
so that Uˆ1 rotates every spin about the x axis by the angle π(1 − εr).
The spin rotation error εr vanishes for the interacting dynamics during the T2 interval, provided ideal pulses.
that heating is suppressed and the time crystalline order For an ideal pulse, εr = 0, the first step reduces to ˆ N ˆ ˆ Q x is preserved.
Fig.2 illustrates this at εr = 0.1.
In the U1 = (−i) P with P = σˆi , and flips every spin, i non-interacting limit, Ji = 0 (with hi = 0), each spin simply rotates by π(1 − ε) per period, and the Fourier ˆ † z ˆ z ˆ †┼ X z◀ ˆ X z r U1 σˆi U1 = −σˆi , U1 σˆi U1 = − σˆi .
(8) peak of the magnetization shifts from ω = π/T to ω = i i π(1 − εr)/T .
In the presence of interactions, Ji ≥ 0, the The stroboscopic magnetization is peak remains sharply locked at ω = π/T .
The interaction term thus counteracts the phase error introduced by the 1 X finite spin-rotation error εr.
It is therefore crucial for both M(nT) = ⟨ψ |(Uˆ †)n σˆz Uˆ n|ψ ⟩.
(9) N 0 F i F 0 the emergence and the stabilization of the subharmonic i response (Else et al., 2016; Khemani et al., 2016; Yao et al., 2017).
Since Hˆ commutes with every σˆz, at ε = 0 we have 2 i r The origin of this locking can be understood from the Uˆ † σˆzUˆ = −σˆz.
This is precisely the dynamical sym-F i F i structure of Uˆ .
At ε = 0, its eigenstates are cat states metry with θ = π introduced in Sec.
IIA and discussed F r |{s}⟩ ± eiϕs |{−s}⟩ built from σˆz product states, with further in Sec.
IIIA. The magnetization therefore alter-quasienergies differing by exactly π/T .
A small rota-nates between successive periods, tion error flips individual spins and therefore creates do-M(nT) = (−1)nM(0),M(n + 2)T ┌ = M(nT).
main walls, which cost an energy of order Ji.
When (10) the couplings and fields are random, these processes are generically off-resonant, so εr acts only perturbatively.
It Thus, the observable responds with period 2T , rather than z z with the drive period T , confirming the subharmonic re-dresses σˆi into quasi-local operators τˆi that still satisfy ˆ † z ˆ z sponse associated with discrete time translation symmetry UF τˆi UF ≃ −τˆi , and the π pairing survives with correc-breaking.
At this fine tuned point, however, the period tions exponentially small in the system size (Else et al., doubling is trivial, since it holds equally in the absence of 2016; Khemani et al., 2016; von Keyserlingk et al., 2016).
interactions. In a genuine discrete time crystalline phase, This is the many-body-localized DTC.
Its rigidity has the response must remain stable against perturbations been established theoretically (Else et al., 2016; Khemani and persist indefinitely in the thermodynamic limit.
et al., 2016; von Keyserlingk et al., 2016; von Keyserlingk The defining feature of the discrete time crystalline & Sondhi, 2016; Yao et al., 2017), and signatures of it have phase is that the subharmonic response remains stable been observed experimentally (Mi et al., 2022; Randall over a finite interval of εr, rather than occurring only at et al., 2021).
the fine-tuned value εr = 0.
This robustness arises from However, quenched disorder is not essential for realizing 5 DTC order, and the asymptotic stability of MBL is itself crystalline order even in one dimension (Pizzi et al., 2021; under debate.
We therefore turn to clean periodically Russomanno et al., 2017).
A prethermal DTC of this kind driven systems, where dynamical many body localiza-has been observed in a trapped-ion chain (Kyprianidis tion (DMBL) can suppress heating through the intrinsic et al., 2021).
dynamics and provide a route toward time crystalline Another route to realizing DTC order in a clean system behavior in the absence of disorder.
A representative ex-is provided by a two-tone drive protocol that dynami-ample is a long-range Ising chain subject to a periodically cally cancels the interaction contribution to the Floquet modulated field, Hamiltonian, resulting in flat quasienergy bands (Baner-jee et al., 2025).
In this scheme, the system is subjected ˆ X y y X z H2(t) = Jij σˆi σˆj − h sin(Ωt) σˆi , to two square wave drives with commensurate frequencies i≤j i ω and ω , satisfying ω = nω , where n ≥ 1 is an odd (11) 1 2 1 2 J integer.
The corresponding time-dependent Hamiltonian J = 0 , ij |i − j|α is ˆ X x x where Ω is the frequency of the field modulation.
As H2(t) = J σˆi σˆi+1 demonstrated by Rahaman et al.(2024a,b), this model i exhibits robust dynamical localization when the

drive h ┌ ┌i X z + h sgn sin ω1t − sgn sin ω2t σˆi , (13) parameters satisfy J0 (4h/Ω) = 0, where J0 denotes the zeroth-order Bessel function of the first kind.
The origin i of this condition is clearest in the frame rotating with the y y ± where sgn denotes the sign function.
For the commen-field.
Writing σˆi σˆj in terms of σˆ , the flip-flop terms surate choice of drive frequencies specified above, the + − P z σˆi σˆj + h.c. conserve i σˆi and are unaffected by the two-tone protocol cancels the interaction contribution + + rotation.
The pair creation terms σˆi σˆj + h.c.
instead over a driving cycle.
The resulting flat quasienergy bands acquire, up to a constant phase, the time dependent phase produce dynamical localization and thereby suppress heat-e±i(4h/Ω) cos Ωt.
Averaged over a cycle, they are multiplied ing.
When combined with a binary Floquet protocol that by J0(4h/Ω) and therefore vanish at its zeros.
To leading includes a spin flip operation, this localization mechanism order in 1/Ω, the effective Hamiltonian then conserves stabilizes a subharmonic response.
If the cancellation the total magnetization, and the resulting dynamical were exact, however, the evolution during T2 would act localization suppresses heating (Rahaman et al., 2024a).
trivially and could not correct spin-rotation errors.
As This localization mechanism can be combined with the in Fig.2, the response would then not be locked.
Small binary Floquet protocol of Eq.
(5), with Hˆ2(t) acting residual interactions restore this rigidity and stabilize the during the interval T2 (chosen as an integer multiple of response against spin-rotation errors, which act as exter-2π/Ω), to stabilize a subharmonic response and thereby nal perturbations.
Rahaman and Roy(2026) showed that realize DTC order in a clean system (Rahaman et al., Floquet flat bands of this kind enable discrete time crys-2024b).
talline order in spin chains, with a robust period doubled A complementary route to realizing DTC order without stroboscopic magnetization.
Thus, this protocol provides disorder is provided by high-frequency driving.
When the a route to realizing clean DTC order without relying on drive frequency ωD is much larger than the local energy quenched disorder.
scale Jloc, absorbing a single drive quantum requires a high-order many-body process.
Consequently, the heating time is exponentially long (Abanin et al., 2017; Kuwahara III.
INTEGRABILITY AND THE (USUAL) et al., 2016; Mori et al., 2016), ABSENCE OF OSCILLATIONS τ exp c ω /J ┌.
(12) heat ≳ D loc An integrable many-body system has an extensive ˆ For times shorter than τ , a suitable rotating frame set of local or quasi-local conserved charges {Qj} with heat ˆ ˆ ˆ ˆ transformation brings the Floquet operator to the form [H, Qj] = [Qj, Ql] = 0 (Essler & Fagotti, 2016; Ilievski −iDTˆ et al., 2016).
Free-fermion models such as the transverse-UˆF ≈ Pˆ e .
The effective static Hamiltonian Dˆ has an ˆ ˆ field Ising chain (Lieb et al., 1961; Pfeuty, 1970) belong to emergent Z2 symmetry, [D, P] = 0.
If the corresponding prethermal state spontaneously breaks this symmetry, this class, as do interacting Bethe-ansatz-solvable models the discrete time translation symmetry is broken and such as the XXZ and Hubbard chains.
Because these the laboratory frame observable exhibits a robust period-charges constrain the dynamics, integrable systems do doubled response (Else et al., 2017).
Thus, the existence of not thermalize.
After a quench they relax locally to a a prethermal DTC is constrained by the same equilibrium generalized Gibbs ensemble (GGE) (Ilievski et al., 2015; symmetry breaking considerations discussed above.
For Rigol et al., 2007; Vidmar & Rigol, 2016), short range interactions, finite temperature breaking of 1 ┼ X ◀ a discrete symmetry requires d ≥ 2.
Clean long range ρˆ = exp − β Qˆ , (14) GGE Z j j interactions can modify this conclusion and support time j 6 where the Lagrange multipliers βj are fixed by the initial the total transverse magnetization precesses at frequency values ⟨Qˆj⟩.

The quantum Newton’s cradle (Kinoshita h without decay.
This does not contradict the Watanabe- ˆ+ et al., 2006) demonstrates this memory of initial condi-Oshikawa theorem: ⟨Stot⟩ vanishes in every Gibbs state tions.
Periodically driven integrable systems similarly of Hˆ , so the oscillation requires a nonequilibrium initial relax to a periodic GGE (Lazarides et al., 2014b).
state. A second example is Yang’s η-pairing operator in Note that integrable systems do relax, just not to a the Hubbard model on a bipartite lattice, which satisfies thermal state.
Expand the state in energy eigenstates, Eq.
(16) with a frequency set by the on-site interaction P |ψ⟩ = m cm |m⟩.
Then and the chemical potential (Yang, 1989).
Medenjak et al.
(2020) argued that isolated Heisenberg magnets with such X 2 ⟨Oˆ(t)⟩ = |cm| Omm structure can be regarded as quantum time crystals.
Long-m lived phantom-helix states, whose longevity reflects exact (15) X ∗ i(E −E)t properties of the anisotropic Heisenberg model, have been + c c O e m n .
m n mn observed with ultracold atoms (Jepsen et al., 2022).
In all m≠n these cases, however, the frequency is set by Hamiltonian In the thermodynamic limit, the energy differences parameters and shifts continuously with them.
Such ∗ Em − En form a continuum.
When the weights cmcnOmn oscillations are therefore not rigid in the sense of criterion are smoothly distributed over it, the second term de- (b).
phases by the Riemann-Lebesgue lemma, leaving only For a periodically driven system, the analogue of the diagonal (GGE) part.
For a periodically driven sys-Eq.
(16) is tem, the same argument applies at stroboscopic times ˆ † ˆ ˆ iθ ˆ t = nT , with energies replaced by quasienergies.
Con-UF A UF = e A, (17) served charges with finite overlap with an observable bound the time average of its autocorrelation function which gives Aˆ(nT) = einθAˆ.
A phase θ = 2π/m produces from below (Mazur, 1969; Suzuki, 1971), but this bound a response with period mT , and θ = π gives period dou-concerns the stationary part of the response rather than bling.
In spectral terms, Aˆ connects Floquet eigenstates oscillations.
Conservation laws therefore determine where whose quasienergies differ by θ/T , which for θ = π is the the system relaxes, but they do not by themselves pre- π pairing of Sec.
IIB. In the kicked Ising chain at εr = 0, z vent relaxation.
Persistent oscillations require more: the every σˆi satisfies Eq.
(17) with θ = π.
Here the phase is spectrum must contain an extensive set of exactly com-not set by a tunable Hamiltonian parameter but by the ˆ2 ˆ mensurate energy differences, and these differences must Z2 structure of the drive, P = 1.
As long as weak per- −iDTˆ contribute finite weight to local observables.
Section IIIA turbations preserve the dressed form UˆF ≈ Pˆ e with explains what algebraic structure gives rise to such com- [D,ˆ Pˆ] = 0, the phase θ = π cannot drift continuously.
mensurability, and Sec.
IIIB discusses its realization in This is the algebraic origin of the rigidity of the period free-fermion models.
in a discrete time crystal, and it distinguishes Floquet dynamical symmetries from their Hamiltonian counter-parts.
Floquet dynamical symmetries of this type have A.
Dynamical symmetries been used to construct time crystals in driven Hubbard models (Chinzei & Ikeda, 2020).
The free-fermion mod-The algebraic structure that produces exactly commen-els of Sec.
IIIB provide a further example, in which the surate energy differences is a dynamical symmetry (Bu˘ca condition is met for selected momenta.
et al., 2019; Medenjak et al., 2020).

An operator Aˆ is a dynamical symmetry of a time-independent Hamiltonian if B.
Free-fermion models [H,ˆ Aˆ] = ωA,ˆ (16) Time-crystalline responses can be realized in a class of ˆ ˆ integrable free-fermion models (Chandra & Roy, 2024), with ω real.
It follows that Aˆ(t) = eiHtAeˆ −iHt = eiωtAˆ, so iωt that ⟨Aˆ(t)⟩ = e ⟨Aˆ(0)⟩ for every initial state, and the os-X (0) Hˆ (t) = Hˆ , cillation never decays.
In spectral terms, if Hˆ |E⟩ = E |E⟩, k k≥0 (18) then Hˆ Aˆ |E⟩ = (E + ω)Aˆ |E⟩.
The operator Aˆ therefore ˆ (0) ┌ ┌ † † ┌ connects eigenstates whose energies differ by exactly ω.
Hk = a0 − bk nk + n−k + Dk ckc−k + h.c.
, When Aˆ is a sum of local terms, it supplies precisely the † extensive set of commensurate energy differences with with n±k = c±kc±k.
This Cooper-pair Hamiltonian is finite weight that Eq.
(15) requires.
specified by the parameters a0, bk, and Dk and by the † The simplest example is the isotropic Heisenberg chain fermionic ladder operators c±k and c±k.
It consists of a ˆ P ˆ ˆ P ˆz in a uniform field, H = J i Si ·Si+1 +h i Si .
Because kinetic term with coupling a0 − bk and a pairing term the exchange term is invariant under global spin rotations, with coupling Dk, where a0 is time dependent.
Consider ˆ ˆ+ ˆ+ the total raising operator satisfies [H, Stot] = h Stot, and driving the system with a square-wave signal of period 7 T and a 50% duty cycle, such that the first half-cycle 2018) showed signatures of time crystalline order with-flips the spins while the second half-cycle does not.
The out engineered disorder.
The mechanism that stabilizes Hamiltonian during the nth cycle is then the response in the dipolar crystal is still debated, with prethermalization and long range interactions as the main (Hˆ (0), nT ≤ t ≤ n + 1 ┌T, candidates.
A prethermal discrete time crystal was sub-Hˆ (t) = k 2 k † † ┌ 1 ┌ sequently demonstrated with trapped ions (Kyprianidis a1 c c + h.c.
, n + T ≤ t ≤ (n + 1)T.
k −k 2 et al., 2021).
Its lifetime grows with the drive frequency, as (19) expected from the slow heating of the prethermal regime For a particular momentum, k = k , if a = b and 0 0 k0 discussed in Sec.IIB.
In Rydberg atom arrays (Bluvstein ω ≡ 2π/T = 2D , the wave function repeats after a D k0 et al., 2021), periodic driving stabilizes period doubled re-time 2T .
This subharmonic response is a hallmark of vivals of quantum many-body scarred states (Serbyn et al., discrete time-crystalline behavior.
2021; Turner et al., 2018).
This experiment shows that For the one-dimensional (1D) transverse-field Ising subharmonic dynamics can arise from restricted, rather model (TFIM), the subharmonic conditions reduce to than complete, algebraic structure in the spectrum.
a0 = bk0 = cos k0 and ωD = 2 sin k0.
However, a small deviation of the momentum from the specific value k0 destroys the time-crystalline response.
In the two-Later experiments addressed eigenstate order directly.
dimensional (2D) TFIM, by contrast, the conditions be-Randall et al.(2021) used a programmable register of come a0 = bk = cos kx + α cos ky and ωD = 2 sin kx + 13C nuclear spins surrounding an NV center to probe 2α sin ky.
Setting α = 0 recovers the 1D case.
The ad-a many body localized discrete time crystal.
Mi et al.
ditional tunable parameter in two dimensions therefore (2022) used a superconducting quantum processor to show permits a continuous range of momenta supporting a that time-crystalline order persists across the spectrum, stable time crystalline response.
In the 1D TFIM, no including for random initial states.
Their echo protocol analogous parameter is available, precluding the construc-was designed to separate the intrinsic decay of the

order tion of a stable time crystal by this mechanism.
from decoherence.
A realization on a large cloud based To obtain stable time-crystalline responses in 1D, one quantum computer followed (Frey & Rachel, 2022), show-can supplement the TFIM with a next-nearest-neighbor ing that such Floquet circuits can now be implemented (NNN), or equivalently three-spin, interaction.
The model on general purpose quantum hardware.
remains integrable, while the subharmonic conditions become a0 = bk = cos k + λ cos 2k and ωD = 2 sin k + 2λ sin 2k, where λ denotes the three-spin coupling.
This Time crystalline behaviour has also been observed be-additional interaction provides the tunability needed to yond closed spin systems.
Magnon condensates in super-construct a 1D time crystal and overcome the limitations fluid 3He exhibit a time quasicrystal and its transition to of the 1D TFIM.
a time crystal (Autti et al., 2018).
In atom cavity systems, dissipation stabilizes a period doubled response (Keßler et al., 2021), and continuous pumping produces self sus-IV.
EXPERIMENTAL REALIZATIONS tained oscillations that break continuous time translation symmetry (Kongkhambut et al., 2022).
These open sys-The first observations of discrete time crystalline order tem realizations show that dissipation can stabilize, rather used two complementary platforms.
Zhang et al.(2017) than only destroy, time crystalline order.
drove a chain of trapped ions with long range Ising inter-actions and engineered disorder.
They observed a period doubled magnetization that remained locked at 2T over Several of these experiments connect directly to the a finite range of pulse imperfections and melted when themes of this chapter.
Floquet circuits on superconduct-the imperfections became large.
Because the chain was ing processors closely match the kicked Ising chain of finite and the interactions long ranged, it is difficult to Eqs.
(5) and (6), with integrability breaking terms that establish that the stabilization in this experiment is gen-can be tuned.
When the longitudinal fields vanish, the uinely due to many body localization.
Choi et al.(2017) circuit reduces to a free fermion model, so the free fermion used a dense ensemble of nitrogen-vacancy (NV) centers physics discussed in Sec.IIIB can be probed directly.
The in diamond with disordered dipolar interactions.
In this phantom helix experiments in ultracold atoms (Jepsen system the order is not stabilized by localization; instead, et al., 2022) probe the dynamical symmetry physics of slow, critical thermalization of the dipolar ensemble ex-Sec.
IIIA in an isolated Heisenberg magnet.
In all plat-tends its lifetime.
Both experiments demonstrated a rigid forms, however, the observation time is limited by deco-subharmonic response, but neither could test persistence herence, and hybrid quantum-classical feedback has been to infinite times.
proposed as a way to prolong discrete time crystalline Experiments in clean systems followed, which is the order on such hardware (Camacho & Fauseweh, 2024).
setting of primary interest in this chapter.
NMR experi-Separating intrinsic persistence, as required by criterion ments on star shaped molecular clusters (Pal et al., 2018) (c), from experimental limitations therefore remains a and on a spatially ordered dipolar crystal (Rovny et al., central challenge.
8 V.
OPEN PROBLEMS AND OUTLOOK der (Gambetta et al., 2019), and in driven atom cavity systems (Keßler et al., 2021; Kongkhambut et al., 2022).
Integrable models give an exact, analytically controlled Algebraic conditions for persistent oscillations and limit view of what a time crystal requires.
They show that cycles under dissipation have been formulated (Bu˘caet al., persistent oscillations need commensurate spectral struc-2022, 2019).

How these open system phases relate to the ture, expressed through dynamical symmetries and the π closed system, eigenstate based definition of a discrete pairing of Floquet eigenstates (Secs.
IIB and IIIA).
In time crystal, and to classical discrete time crystals (Zaletel the one-dimensional TFIM this structure is confined to et al., 2023), remains to be clarified.
isolated momenta (Sec.
IIIB). Interactions are needed to Responses beyond period doubling also deserve at-lock the response and to stabilize it against the domain tention.
Higher-order and fractional subharmonic re-walls created by pulse errors (Sec.IIB).
Integrable models sponses (Pizzi et al., 2021), as well as time quasicrys-thus explain why the structure required for a time crystal tals (Autti et al., 2018), remain largely unexplored in is generically fragile.
Several directions remain open.
integrable settings, where exact solutions could clarify The first concerns clean interacting systems.
It is not their spectral origin.
Equation (17) with θ = 2π/m pro-known whether there exists a rigorous, non perturbative vides a natural starting point for such an analysis.
example of a bulk discrete time crystal in a clean, short Finally, the asymptotic status of many body localization range, one dimensional system that persists indefinitely.
remains unsettled.
Although MBL has been proven in one dimension under an assumption on level statistics (Imbrie, Prethermal mechanisms cannot provide such an exam- ˇ ple, since a discrete symmetry cannot be spontaneously 2016), numerical studies questioning its stability (Suntajs broken at finite temperature in one dimension with short et al., 2020) and the identification of avalanche instabil-range interactions.
Edge modes offer an alternative at ities (De Roeck & Huveneers, 2017; Morningstar et al., the boundary.
Floquet Majorana π modes produce per-2022) have reopened the question.
Its resolution will de-sistent period doubling at the ends of driven free fermion termine whether disordered discrete time crystals are true chains (Thakurathi et al., 2013), and strong and almost phases of matter or merely extremely long lived crossovers.
strong zero modes give long lived edge spins in interacting This question also motivates the disorder free mechanisms chains (Fendley, 2016; Kemp et al., 2017).
Extending emphasized in this review.
such edge modes to interacting Floquet chains provides Time crystals extend the concept of spontaneous sym-one route toward long lived boundary oscillations.
In the metry breaking to systems far from equilibrium.
Their bulk, clean long range interactions in one dimension (Pizzi defining feature is not oscillation itself but its rigidity, et al., 2021; Russomanno et al., 2017), or short range in-which in Floquet systems originates from a discrete sym-teractions in higher dimensions (Else et al., 2017), allow metry of the drive that pins the response at a fixed frac-time crystalline order without disorder.
tion of the drive frequency.
Integrable models show most A second direction concerns the robustness of dynamical clearly what this requires, namely commensurate spec-symmetries.
The dynamical symmetries of Heisenberg and tral structure protected against perturbations, and why Hubbard models discussed in Sec.IIIA produce persistent generic systems struggle to provide it.
Whether clean in-oscillations, but their frequencies are set by Hamiltonian teracting systems can provide it indefinitely, rather than parameters and shift continuously with them.
They are for exponentially long but finite times, remains the central therefore not rigid in the sense of criterion (b).
An impor-open question of the field.
tant question is whether such symmetries can be made robust, for instance through drive-induced locking, so
ACKNOWLEDGMENTS that their frequencies become rigid.
The contrast with the Z2 structure of the kicked Ising chain suggests that a discrete symmetry of the drive is the essential

ingredient.
SM thanks the University Grants Commission, Gov-A third direction lies between the integrable and chaotic ernment of India, for financial assistance through the limits.
Quantum many body scars (Serbyn et al., 2021; NET fellowship.
The authors acknowledge Claude (Opus Turner et al., 2018) and Hilbert space fragmentation (Sala 5.5, Anthropic) and Copilot for consistency checks.
The et al., 2020) generate long lived oscillations from restricted authors reviewed all AI-assisted changes and take full algebraic structure.
In scarred systems, approximately responsibility for the manuscript.
equally spaced towers of eigenstates play the role of a dynamical symmetry acting within a subspace, and peri-REFERENCES odic driving can stabilize the resulting revivals (Bluvstein et al., 2021).
This similarity suggests that a single alge-braic framework may describe non stationary many body Abanin, D., et al.
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quantum systems.
Communications in Mathematical Open systems raise a further set of questions.
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